Artificial Intelligence and Cybersecurity

Overview

3 CFU

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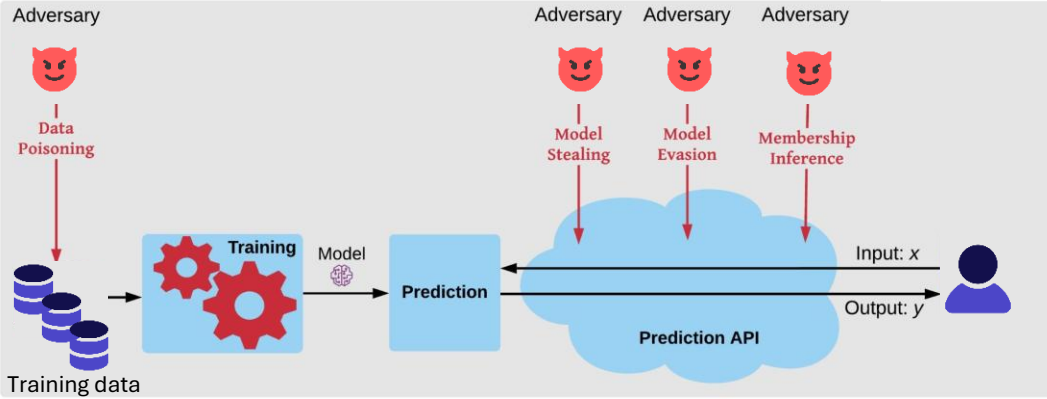
Security **of** ML Systems

First 3 CFU

- **Security: training data poisoning, model evasion, model stealing, jailbreak**
- **Privacy: membership inference, privacy leakage auditing**



Lecture Goals

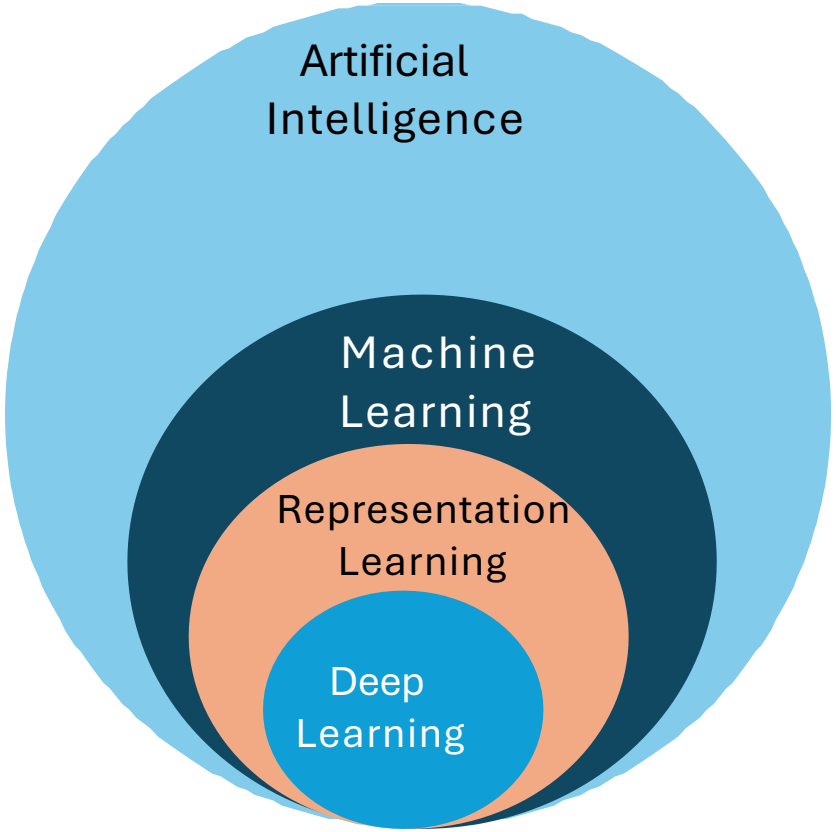
- Logistics:
  - Course objectives, topics
- Motivation and Overview:
  - AI/ML progress
  - AI/ML pitfalls and risks
  - The broader area of Trustworthy AI/ML

Course Objectives

- ML in adversarial environments: security, privacy and countermeasures
- ML safety risks and countermeasures
- ML transparency and interpretability

Where are we

Modern AI/ML in Context



Key Enablers of Recent ML Boom

SGD

…1952

1958

Perceptrons:
Learnable weights

Back propagation

Multi-layer Perceptron

1986

1995

Deep ConvNets
Handwritten digit recognition

Explosive ML boom

Early 2000's …

The big data revolution: Internet-scale data collection & storage

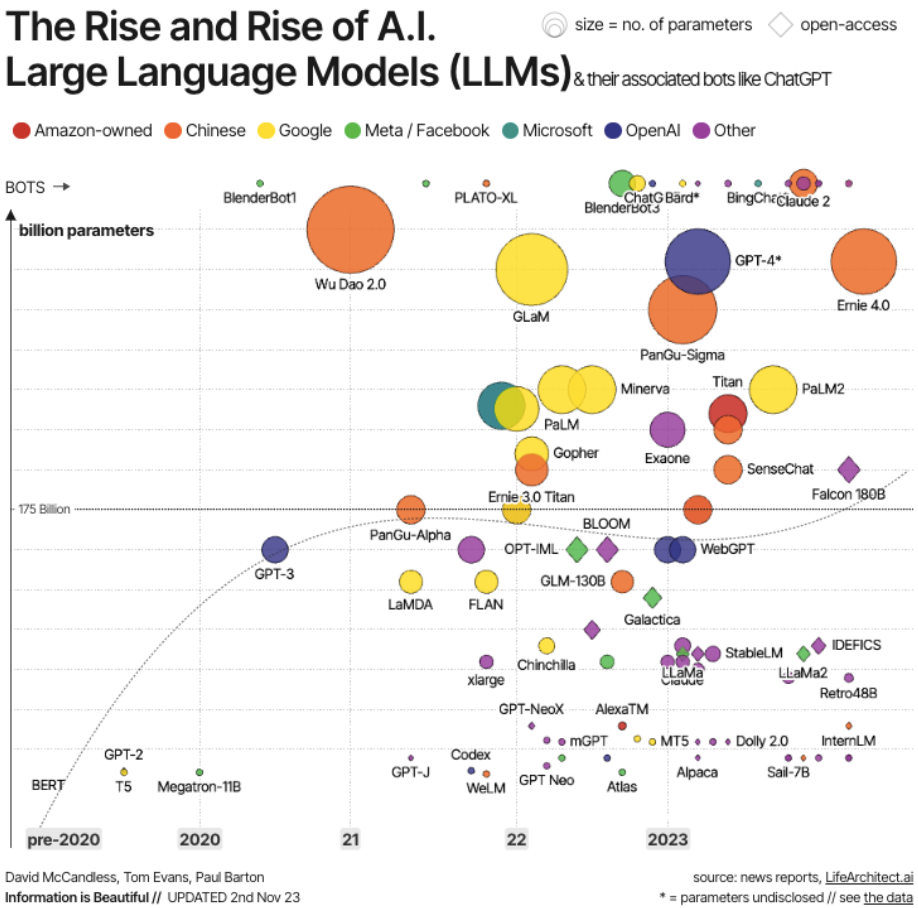
Hardware acceleration and cost reduction: GPUs, TPUs, parallelization, …

Advances in algorithms: techniques, libraries, …

TensorFlow

PYTORCH

Growth of Language Model Size



Src: https://www.linkedin.com/pulse/guide-large-language-models-how-work-yugank-aman-g1wif/

Traditional ML

Let’s discuss it together

Typical Machine Learning Setting

- Task: supervised learning
- Model : classifier (e.g., spam, malware, image)

And what does enable us to do?

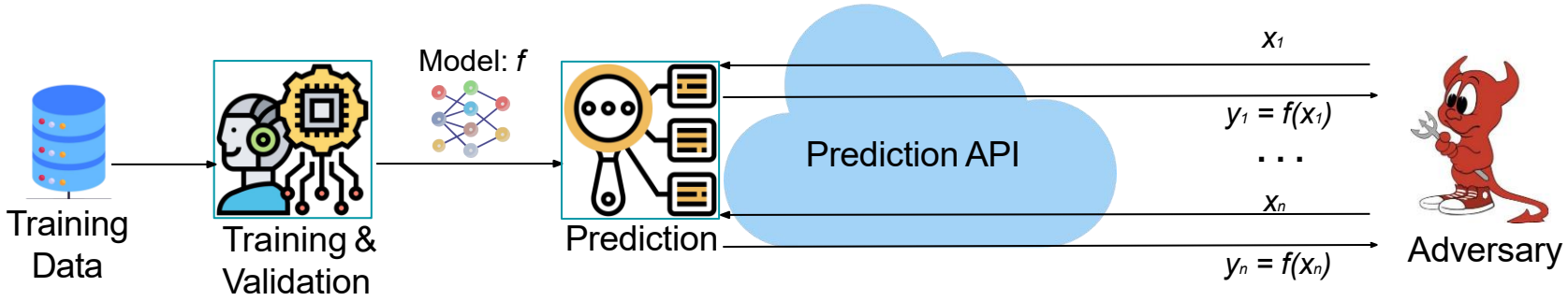
What can go wrong?

General framework in which we will move

ML in the Presence of Adversaries

- We will explore what happens if we replace the user with an adversary

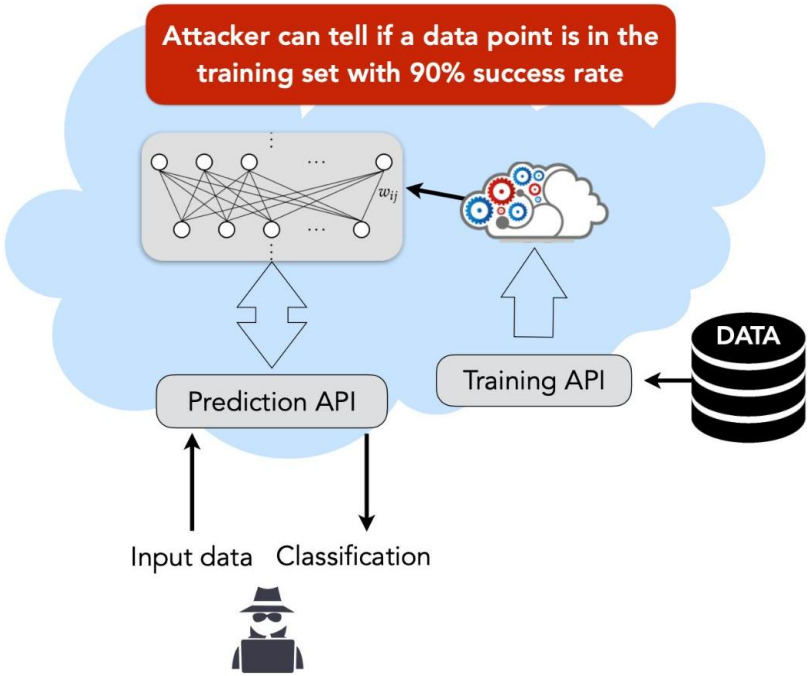
Privacy: Membership Inference Attack



Adversarial Goal: inspect a model's predictions to determine whether or not a sample was used to train it.

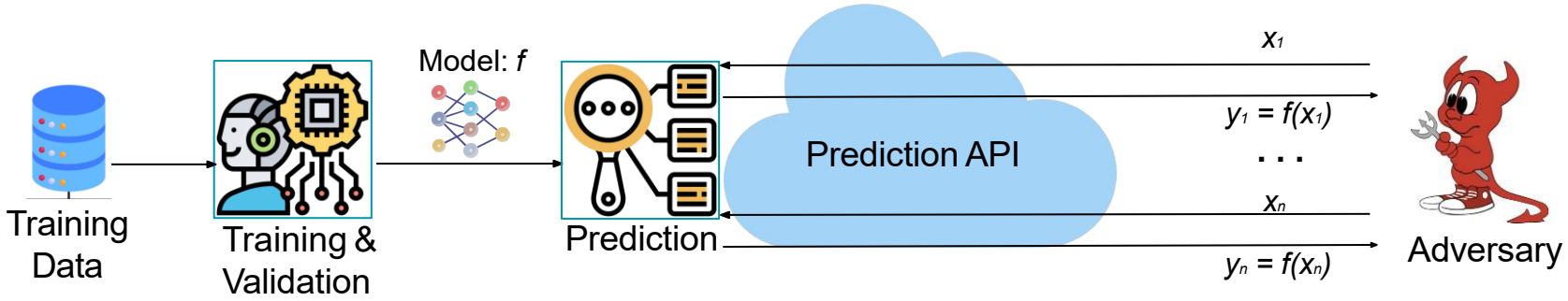
Shokri *et al*. https://arxiv.org/abs/1610.05820; Choquette-Choo *et al.* https://arxiv.org/abs/2007.14321v2

Privacy: Membership Inference …



Shokri *et al*. https://arxiv.org/abs/1610.05820

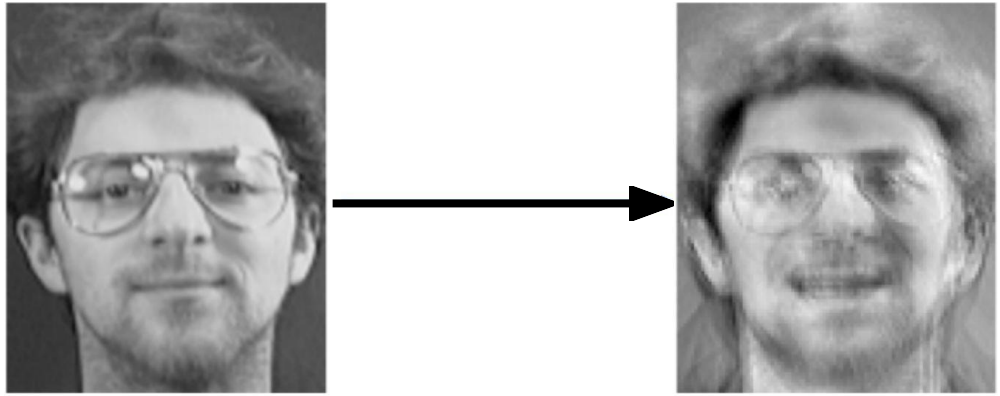
Privacy: Model Inversion Attack



Adversarial Goal: given a model's prediction (e.g., with confidence scores), reconstruct a sensitive training data-point (e.g., a person's face, a medical record)

Fredrikson *et al.*, 2015. Model Inversion Attacks that Exploit Confidence Information and Basic Countermeasures

Privacy: Model Inversion Attack …



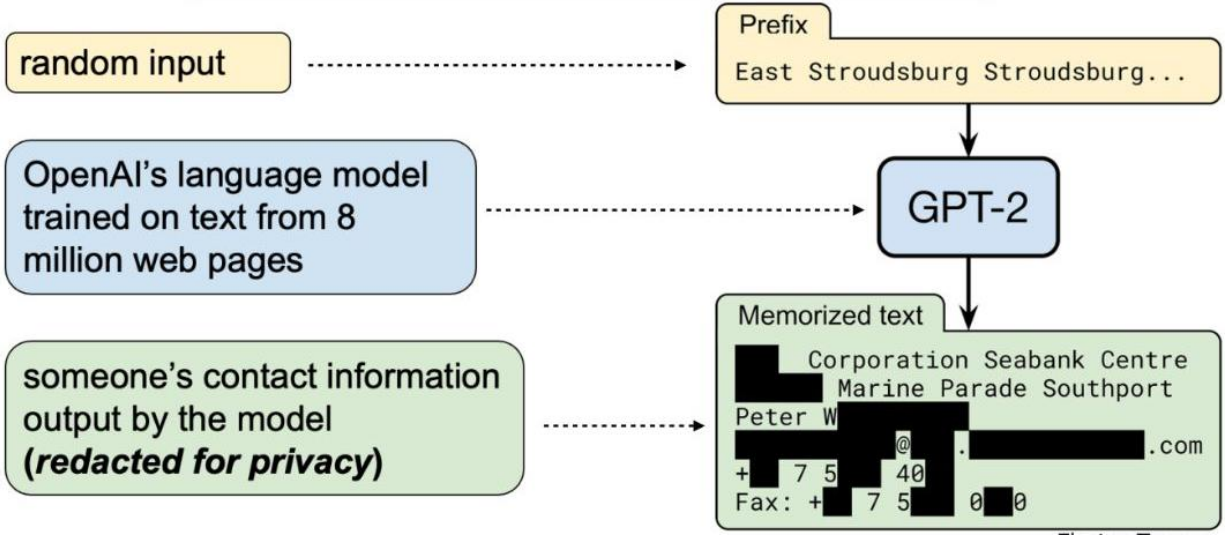
Original face from training data

Reconstructed face after inversion attack

"The attacker is given only the person's name and access to a facial recognition system that returns a class confidence score"

Fredrikson *et al.*, 2015. Model Inversion Attacks that Exploit Confidence Information and Basic Countermeasures

Privacy: Training Data Extraction from LLMs

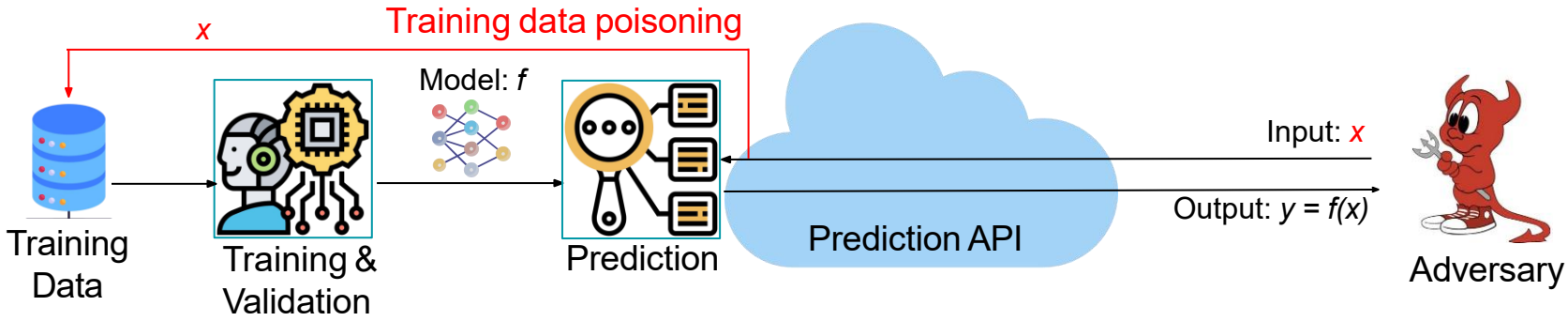


"Given query access to a neural network language model, we extract an individual person's name, email address, phone number, fax number, and physical address. The example in this figure shows information that is all accurate so we redact it to protect privacy."

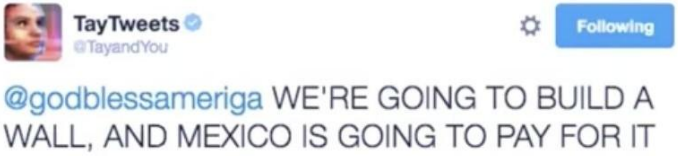
Carlini *et al.*, 2021. Extracting Training Data From Large Language Models

Security and Safety

Security: Data Poisoning Attack



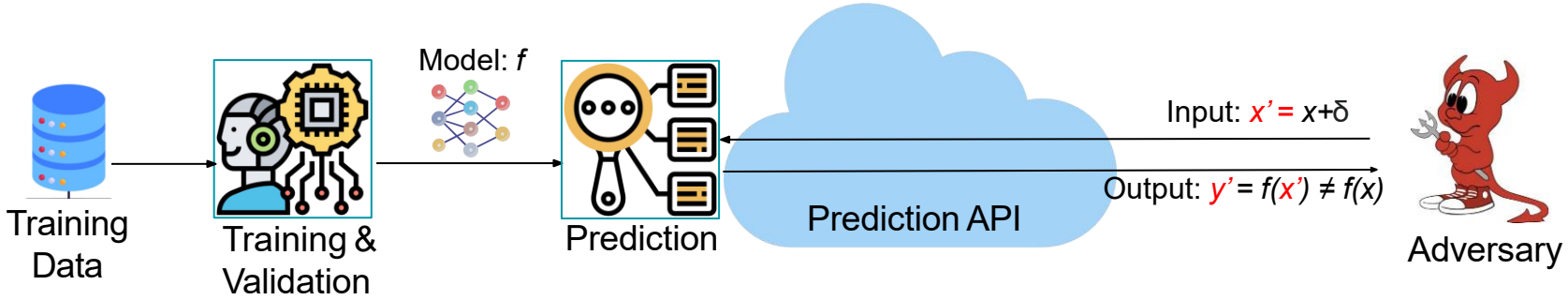
Adversarial Goal: inject training samples that skew model's decisions to advance adversarial intent.



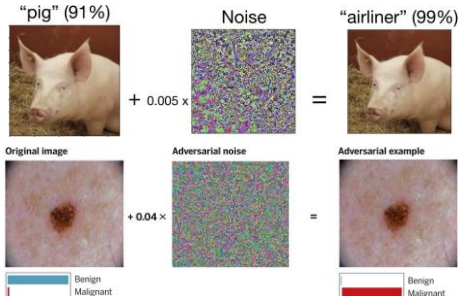
Label: 2 Label: 5 Label: 9 Label: 4 **Label: 5**

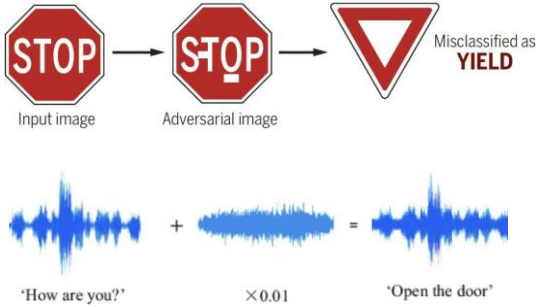
Jagielski *et al*. https://arxiv.org/abs/1804.00308

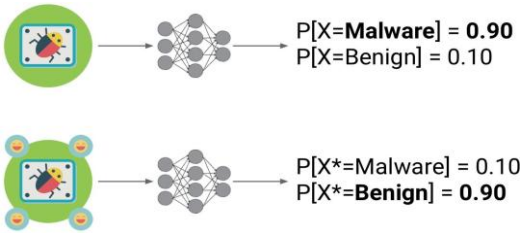
Security: Adversarial Example Attack



Adversarial Goal: craft an adversarial example that forces the model to make incorrect predictions.

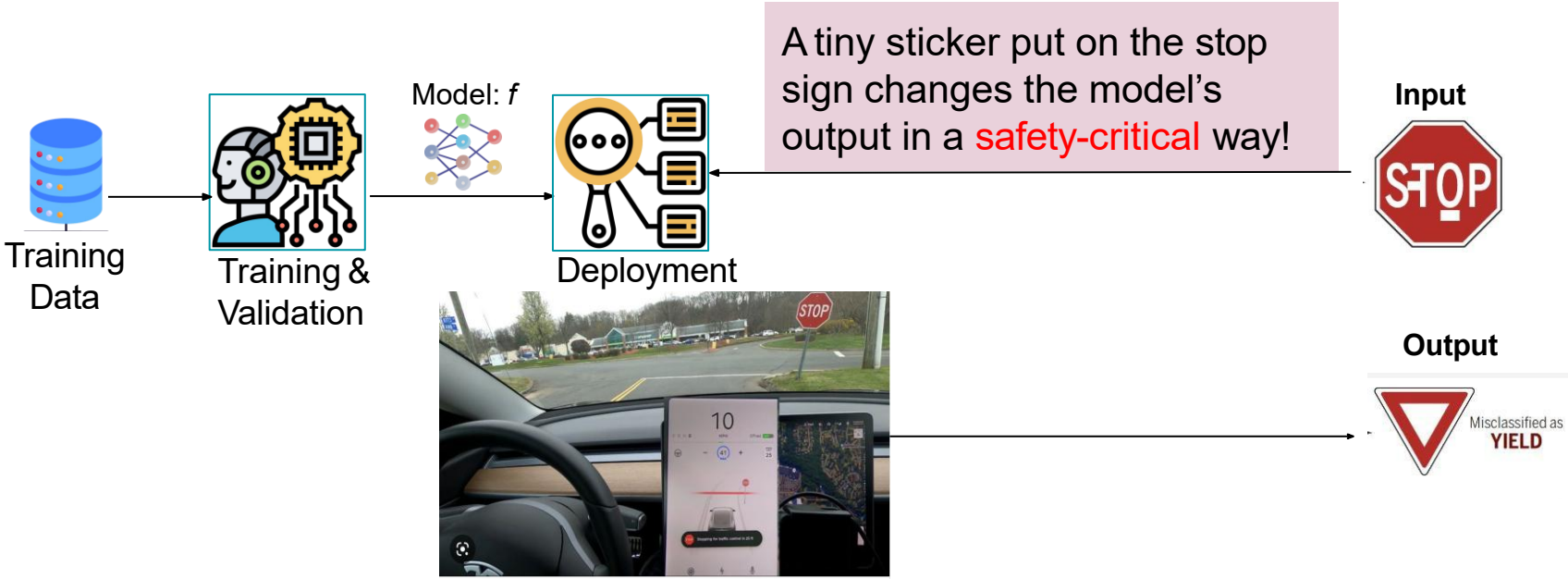






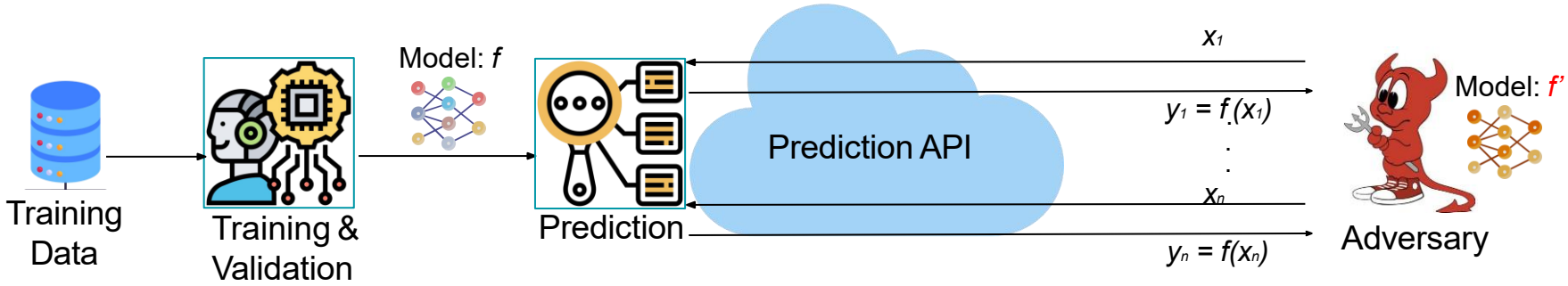
Szegedy et al. https://arxiv.org/abs/1312.6199; Finlayson et al. https://www.science.org/doi/10.1126/science.aaw4399; Eykholt et al. https://arxiv.org/abs/1707.08945; Grosse et al. https://arxiv.org/pdf/1606.04435.pdf

Safety: Traffic Sign Mis-Detection

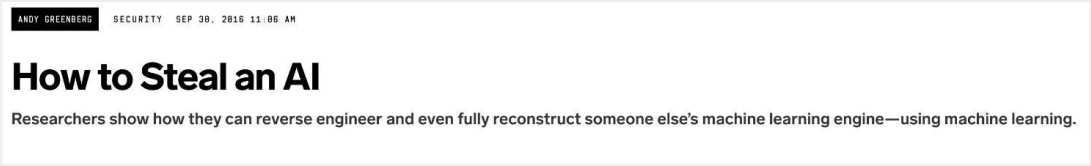


Would you trust the self-driving car or would you rather take over the wheel?

Security: Model Stealing/Extraction



Adversarial Goal: use target model *f* as an oracle to locally train a substitute model *f'* such that *f'* ≈ *f*.


ANDY GREENBERG SECURITY SEP 30, 2016 11:06 AM

How to Steal an AI

Researchers show how they can reverse engineer and even fully reconstruct someone else's machine learning engine—using machine learning.

Tram`er *et al*. https://arxiv.org/abs/1609.02943; Jagielski *et al.* https://arxiv.org/abs/1909.01838; Ali and Eshete. https://arxiv.org/abs/2006.15725

Multifaceted Risks Posed by LLMs

Discrimination, exclusion and toxicity

Harms that arise from the language model producing discriminatory and exclusionary speech.

Information hazards

Harms that arise from the language model leaking or inferring true sensitive information.

Misinformation harms

Harms that arise from the language model producing false or misleading information.

Malicious uses

Harms that arise from actors using the language model to intentionally cause harm.

Human-computer interaction harms

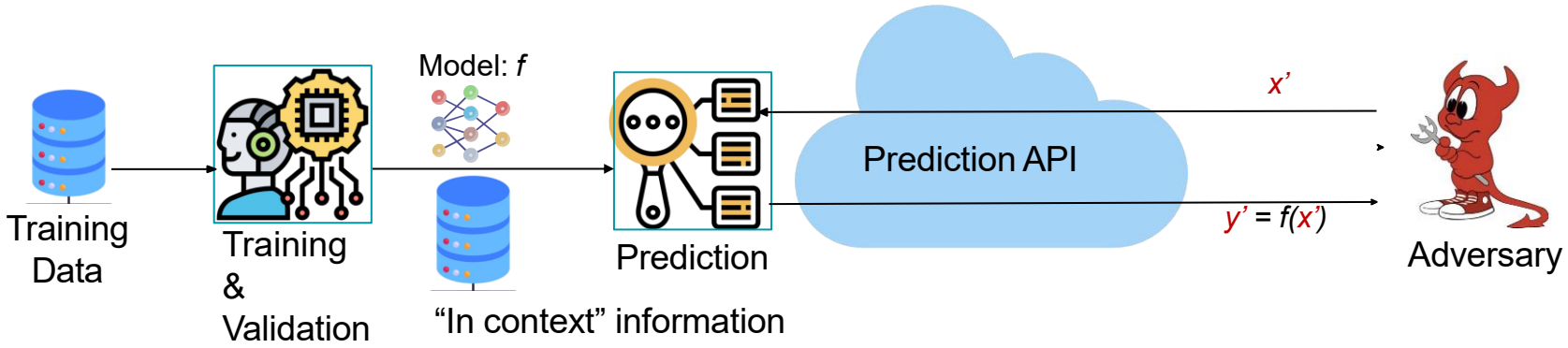
Harms that arise from users overly trusting the language model, or treating it as human-like.

Automation, access and environmental harms

Harms that arise from environmental or downstream economic impacts of the language model.

Weidinger *et al.* 2021. Taxonomy of Risks posed by Language Models

Security: Jailbreak and Prompt Injections

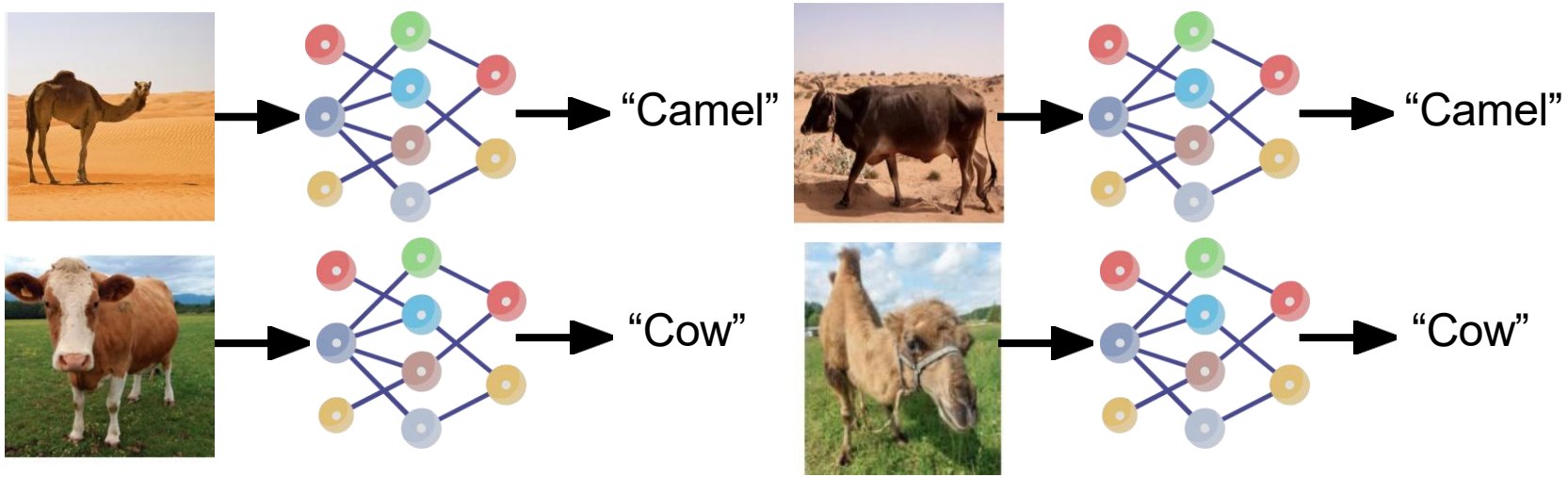


Adversarial Goal: craft adversarial inputs to use target model *f* in inappropriate ways, violating policies or exfiltrating sensitive data used at **inference** time

TODO

Transparency and Interpretability

Interpretability: Cow vs. Camel Classifier



In the absence of explanations for decisions, it is hard to understand the how/why of model behavior.

The Curse of I.I.D. Generalization

Training Environment

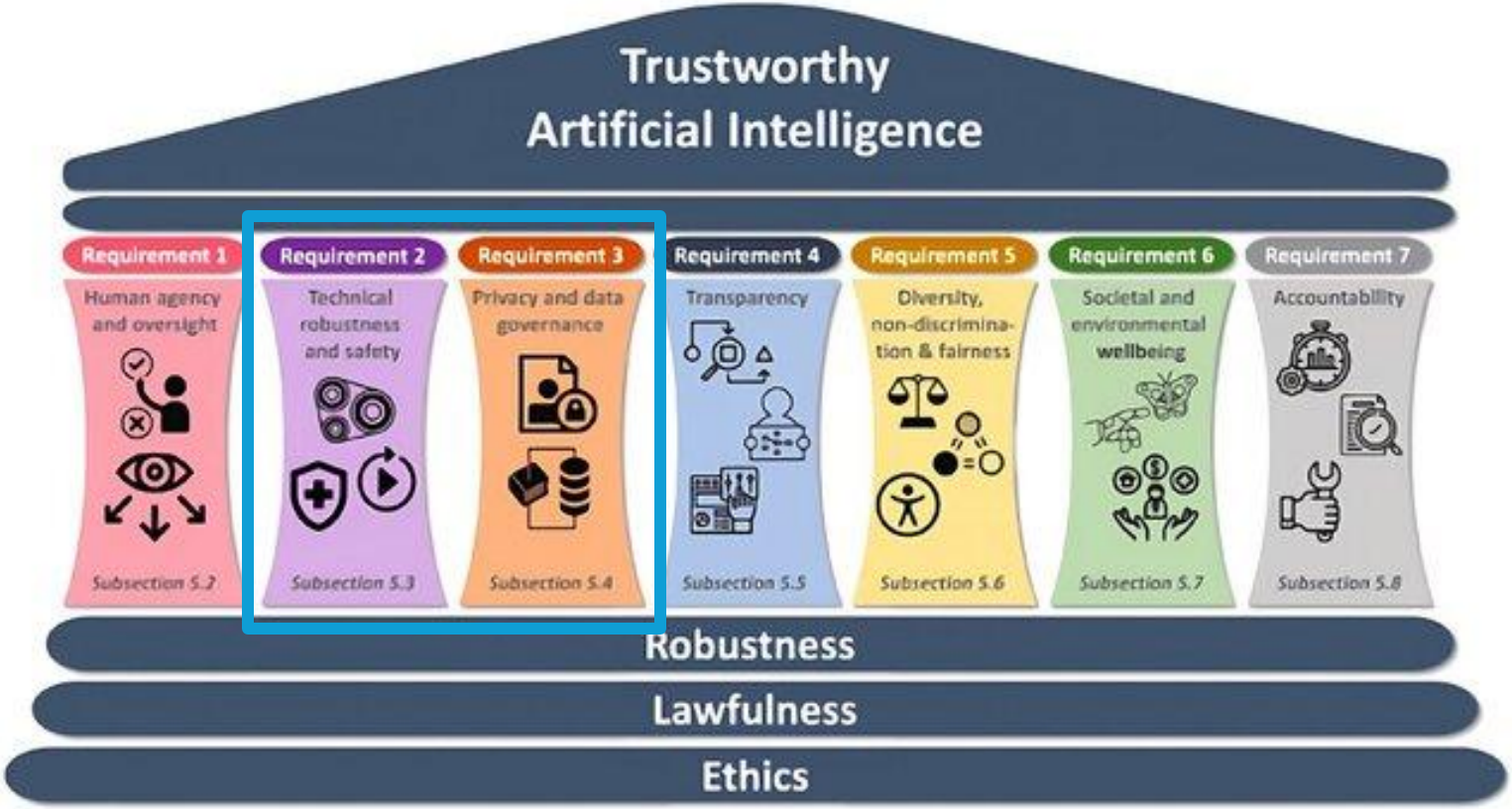
Deployment Environment

ML models struggle to maintain their accuracy when deployed in practice, specifically, when facing inputs that lie *out of distribution* of their train/test data.

Amich and Eshete, 2021. [Rethinking Machine Learning Robustness via its Link with the Out-of-Distribution Problem]

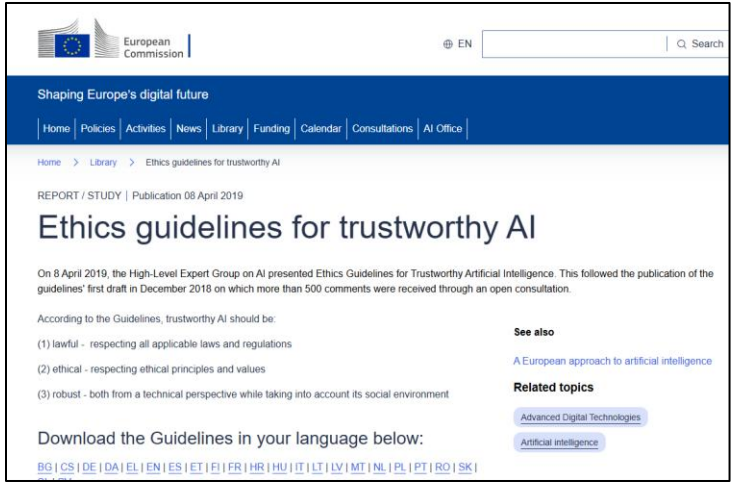
Trustworthy ML

A broader framework

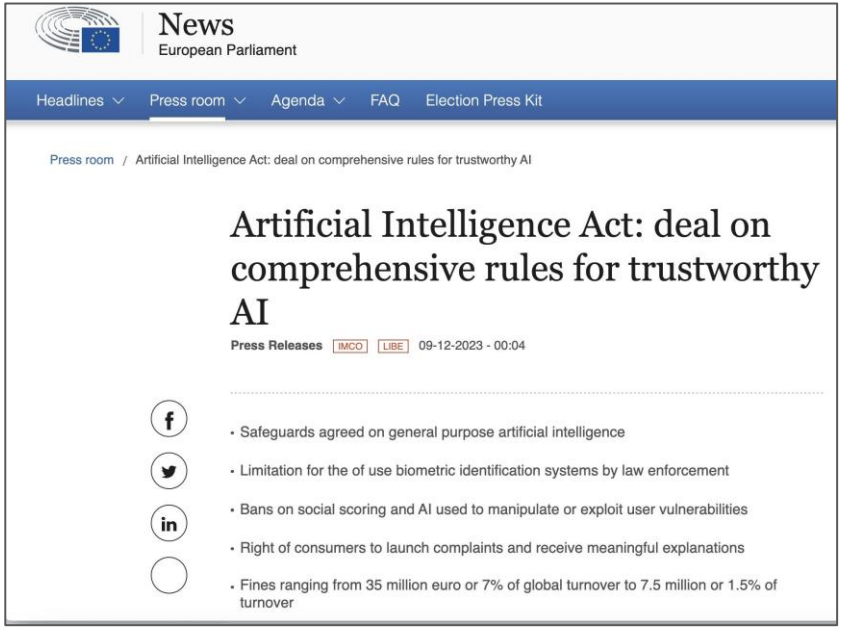


Src: Artificial Intelligence and Digital Twins for Bioclimatic Building Design: Innovations in Sustainability and Efficiency

Regulatory Frameworks: Europe



https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai



https://www.europarl.europa.eu/news/en/press-room/20231206IPR15699/artificial-intelligence-act-deal-on-comprehensive-rules-for-trustworthy-ai

Acknowledgments

- Slide Content
  - Adapted from https://trustworthy-ml-course.github.io/ Lecture 1

- Icons
  - https://www.flaticon.com/authors/saepul-nahwan